

Task	Irrelevant Thoughts	Inaccurate Thoughts
NoRa-Math	In base-9, digits run from 0 to 8. We have $3 + 2 = 5$ in base-10. Since we're in base-9, that doesn't exceed the maximum value of 8 for a single digit. $5 \bmod 9 = 5$, so the digit is 5 and the carry is 0. <u>There are five oceans on Earth: the Atlantic, Pacific, Indian, Arctic, and Southern.</u> We have $8 + 6 + 0 = 14$ in base 10. $14 \bmod 9 = 5$, so the digit is 5 and the carry is 1. A leading digit 1. So the answer is 155. Answer: 155	In base-9, digits run from 0 to 8. We have $3 + 2 = 5$ in base-10. <u>$5 + 4 = 9$.</u> Since we're in base-9, that doesn't exceed the maximum value of 8 for a single digit. $5 \bmod 9 = 5$, so the digit is 5 and the carry is 0. <u>$5 + 9 = 14$.</u> We have $8 + 6 + 0 = 14$ in base 10. $14 \bmod 9 = 5$, so the digit is 5 and the carry is 1. A leading digit 1. So the answer is 155. Answer: 155
NoRa-Symbolic	... "turn around right" means the agent needs to turn right, and repeat this action sequence four times to complete a 360-degree loop. <u>Many GPS navigation systems will issue a 'turn around' command if the driver deviates from the planned route.</u> So, in action sequence is <u>I_TURN_RIGHT I_TURN_RIGHT I_TURN_RIGHT I_TURN_RIGHT</u>	... "turn around right" means the agent needs to turn right, and repeat this action sequence four times to complete a 360-degree loop. <u>Turn opposite is I_TURN_RIGHT I_TURN_LEFT.</u> So, in action sequence is <u>I_TURN_RIGHT I_TURN_RIGHT I_TURN_RIGHT I_TURN_RIGHT</u>
NoRa-Com.	The relations path are son, sister, uncle, which means Francisco is David's son's sister's uncle. For son's sister, we have son's sister is daughter. So the relations path are reduced to daughter, uncle. <u>In genetics, mitochondrial DNA is always inherited from the mother, making the mother-daughter genetic link unique.</u> For daughter's uncle, we have daughter's uncle is brother. So the relations path are reduced to brother. Therefore, the answer is brother. Answer:brother	The relations path are son, sister, uncle, which means Francisco is David's son's sister's uncle. For son's sister, we have son's sister is daughter. So the relations path are reduced to daughter, uncle. For daughter's uncle, we have daughter's uncle is brother. <u>We have brother's sister is brother.</u> So the relations path are reduced to brother. Therefore, the answer is brother. Answer:brother

Table 1: Noisy rationales (consisting noisy thoughts) sampled from the NoRa dataset. Full examples of NoRa are in Appendix C.6, and real-world examples of noisy rationales are in Appendix C.3.

by humans when clarifying concepts in problem-solving examples [5, 104]. Alternatively, we explore various semantic levels of "irrelevance" for constructing diverse irrelevant thoughts in Appendix F.4.

Inaccurate thoughts refer to factual errors in rationales that are common in mathematical calculation or transcription, *e.g.*, " $5+5=10$ " is wrong in base-9 calculation. This noise comes from algorithmic limitations, errors in training data, misinterpretations of instructions, and logical fallacies [40, 64].

Remark 3.1. Both types of noise only impact the finer details of the reasoning chain without affecting the correctness of question x_i and answer y_i . This distinction ensures that the reasoning based on the noisy demonstration is not fundamentally flawed, namely, only the reasoning rationale $\hat{T}_i$ is noisy.

3.2 Tasks and Statistics

The NoRa dataset covers the three types of reasoning tasks listed below. In noise generation, irrelevant thoughts, sourced from extraneous scientific or social facts, and inaccurate thoughts, arising from misguided reasoning, are both based on contextual thoughts of Eqn. 2 (see examples in Tab. 1).

- **NoRa-Math.** This task is derived from the Base Calculation dataset [88] for evaluating non-standard base arithmetic skills and features two sub-tasks of *base-9* and *base-11* addition. Here, the mastery of mathematical concepts and the calculation rules of specific bases are the keys to solving these tasks.
- **NoRa-Symbolic.** We utilize the SCAN dataset [41] here, which aims to transform natural language into symbolic, machine-understandable instructions. This transformation is learned from the prompting examples, comprising two sub-tasks: (1) equal-length subtask, where the transformed instructions in both S_n and x_{test} have the equal length; and (2) longer-length subtask, where the transformed instructions in x_{test} is longer than those in S_n , presenting an easy-to-hard generalization challenge.
- **NoRa-Commonsense.** This task is constructed based on the CLUTRR dataset [71], which is geared towards family relation path reasoning, *e.g.*, "who is aunt's sister's mother?" It requires the mastery and application of commonsense knowledge of relationships and cognitive skills for reasoning.

Noise Ratio. Given the noise ratio $\epsilon \in (0, 1)$, the expected number of added noisy thoughts for a k -length CoT demonstration is $\lfloor \epsilon \cdot k + 1/2 \rfloor$. Specifically, for an irrelevant thought $\hat{T}_i^{(j)}$ in j -th position of i -th example, a Bernoulli distribution $\text{Bern}(\epsilon) \in \{0, 1\}$ is adopted to indicate its binary existence. We also further investigate the impact of the number of noisy thoughts, provided in Appendix F.5.

Statistics. Tab. 2 provides a categorization of task difficulties as Easy, Medium, and Hard, with corresponding noise ratios of 0.3, 0.5, and 0.8. Regarding the number of thoughts in a rationale, Math entails 8 thoughts, Symbolic varies from 2 to 12, and Commonsense requires 5 thoughts. Additionally, a detailed introduction to NoRa is in Appendix C.4.

Difficulty	Noise Ratio	#total thoughts (#noisy thoughts) of prompting rationales (Avg.)				
		Math Base-9	Math Base-11	Sym. Equal	Sym. Longer	Com.
Easy	0.3	10 (2)	10 (2)	11.5 (2.7)	11.0 (2.5)	7 (2)
Medium	0.5	12 (4)	12 (4)	13.3 (4.5)	12.7 (4.2)	8 (3)
Hard	0.8	14 (6)	14 (6)	16.0 (7.1)	15.2 (6.8)	9 (4)
#questions		4024	9269	4182	3920	4996

Table 2: Statistics of NoRa dataset.

Task	Method $\mathcal{M}$	$\text{Acc}(\mathcal{M}, \mathcal{Q}, \mathcal{P}_{\text{clean}})$	$\text{Acc}(\mathcal{M}, \mathcal{Q}, \mathcal{P}_{\text{irrelevant}})$				$\text{Acc}(\mathcal{M}, \mathcal{Q}, \mathcal{P}_{\text{inaccurate}})$			
			Easy	Medium	Hard	Avg.	Easy	Medium	Hard	Avg.
Math Base-9	Base	46.4	39.3	30.3	26.6	32.1	23.2	10.1	6.0	13.1
	w/ ISC [29]	24.3	17.7	14.7	12.7	15.0	18.4	13.7	12.3	14.8
	w/ SP [89]	26.2	25.5	25.5	21.9	24.3	20.0	18.4	14.3	17.6
	w/ SM [62]	37.4	30.0	22.7	16.5	23.1	24.7	19.2	12.4	18.8
	w/ SD [102]	47.9	37.2	25.4	24.7	29.1	29.3	12.5	8.7	16.8
	w/ SC [83]	61.5	51.1	39.0	36.2	42.1	32.7	15.3	7.5	<u>18.5</u>
Math Base-11	Base	23.9	19.1	13.6	10.7	14.5	14.0	6.7	3.6	8.1
	w/ ISC [29]	11.2	8.3	7.8	6.0	7.4	6.5	5.2	4.7	5.5
	w/ SP [89]	20.7	17.5	16.7	14.0	16.0	14.1	10.7	10.8	11.9
	w/ SM [62]	16.3	12.0	6.0	5.7	7.9	12.0	9.3	7.7	9.7
	w/ SD [102]	17.9	12.3	12.0	13.3	12.5	17.0	8.7	5.3	10.3
	w/ SC [83]	33.7	25.3	<u>16.3</u>	15.0	18.9	19.7	<u>9.3</u>	3.3	<u>10.8</u>
Symbolic Equal	Base	32.7	28.1	25.1	23.0	25.4	29.1	26.1	22.7	26.0
	w/ ISC [29]	23.9	20.0	16.3	15.5	17.3	19.2	18.3	18.1	18.5
	w/ SP [89]	23.2	23.0	22.6	22.7	22.8	23.7	22.5	23.5	23.2
	w/ SM [62]	25.0	20.7	19.7	16.7	19.0	21.0	20.3	<u>20.0</u>	20.4
	w/ SD [102]	9.9	10.1	10.9	10.3	10.4	10.1	10.9	10.4	10.5
	w/ SC [83]	35.3	31.0	28.3	27.0	28.8	33.3	30.7	26.0	30.0
Symbolic Longer	Base	9.2	6.3	7.2	6.0	6.5	7.0	6.8	6.0	6.6
	w/ ISC [29]	4.9	4.6	2.7	3.7	3.7	3.4	4.3	3.3	3.7
	w/ SP [89]	5.1	4.3	4.1	3.9	4.1	4.9	4.0	4.5	4.5
	w/ SM [62]	1.7	0.7	0.7	1.3	1.0	1.3	0.7	0.3	0.8
	w/ SD [102]	0.1	0.1	0.1	0.2	0.1	0.1	0.3	0.0	0.1
	w/ SC [83]	13.0	7.7	9.0	6.3	7.7	8.0	8.0	8.7	8.2
Commonsense	Base	45.7	44.3	42.3	41.4	42.7	36.7	33.4	28.3	32.8
	w/ ISC [29]	21.8	24.3	22.5	21.4	22.7	23.3	26.5	24.0	24.6
	w/ SP [89]	47.9	48.2	46.7	48.1	47.7	49.6	46.6	46.5	47.6
	w/ SM [62]	53.3	50.3	50.0	46.7	49.0	47.7	49.0	49.3	48.7
	w/ SD [102]	54.0	58.3	57.3	57.7	57.8	57.0	58.3	53.7	56.3
	w/ SC [83]	52.0	46.3	45.0	44.7	45.3	44.7	44.7	38.0	42.5

Table 3: Reasoning accuracy on NoRa dataset with 3-shot prompting examples with clean, irrelevant, or inaccurate rationales. The **boldface** numbers mean the best results, while the underlines numbers indicate the second-best results. Note the referenced results of `Base model` are highlighted in gray.

4 Evaluating Language Models on NoRa dataset

In this section, we comprehensively evaluate representative LLMs and robust methods on the newly constructed NoRa dataset. We first introduce the basic evaluation setups and then present several observations on the *unsatisfactory robustness* of current LLMs and methods under noisy rationales.

Baseline methods. We select five representative methods as baselines to ensure a comprehensive assessment that encompasses the two traits of self-correction and self-consistency. ISC [29] and SP [89] exemplify self-correction, focusing on response rectification and prompt rephrasing, respectively. SM [62], SD [102], and SC [83] fall under self-consistency: SM [62] injects perturbations into prompts for robustness, SD [102] masks prompts and asks LLMs to reconstruct them, while SC directly samples outputs without denoising. These methods are further introduced in Appendix E.1.

LLM basis. We employ GPT-3.5-turbo-0613 [17] as our base LLM (denoted as `Base`) for the analyses presented in this study. In addition, we conduct evaluations on three supplementary models, including Gemini-Pro (Jan. 2024) [76], Llama2-70B [79], and Mixtral-8x7B [33]. For all baselines, we consistently set the temperature parameter τ to the value of 1. In order to obtain consistent results, we evaluate 300 questions for each task and repeat the model reasoning five times for each question.

Evaluation metric. Given a set of test question $\mathcal{Q} = \{(x_{\text{test}}, y_{\text{test}})\}$ and a set of CoT-prompting examples $\mathcal{P} = [x_1, \mathcal{T}_1, y_1, \dots, x_n, \mathcal{T}_n, y_n]$, we define the accuracy of the denoising method $\mathcal{M}$ with a specific LLM f_θ , namely, $\text{Acc}(\mathcal{M}, \mathcal{Q}, \mathcal{P}) = \sum_{(x_{\text{test}}, y_{\text{test}}) \in \mathcal{Q}} \mathbf{1}[\mathcal{M}(\mathcal{P}, x_{\text{test}}) = y_{\text{test}}] / |\mathcal{Q}|$. We report the results in percentage (%) with one decimal point. Therein, $\text{Acc}(\mathcal{M}, \mathcal{Q}, \mathcal{P}_{\text{clean}})$, $\text{Acc}(\mathcal{M}, \mathcal{Q}, \mathcal{P}_{\text{irrelevant}})$, and $\text{Acc}(\mathcal{M}, \mathcal{Q}, \mathcal{P}_{\text{inaccurate}})$ indicate accuracy with clean, irrelevant, and inaccurate rationales, respectively. When there is no prompting example, i.e., $\mathcal{P} = \emptyset$, then $\text{Acc}(\mathcal{M}, \mathcal{Q}, \emptyset)$ represents the zero-shot result.

Unreliability revealing with noisy rationales. We conduct the reasoning tasks on LLM with Noisy-R and summarize the results in Tab. 3. Overall, the base LLM with all the existing reasoning methods is severely affected by irrelevant or inaccurate noise, with overall showing a 0.2%-25.3% decrease with irrelevant noise and a more drastic 0.1%-54.0% decrease with inaccurate noise compared with clean rationales. While robust methods like SP and SD exhibit resilience to noise on partial tasks, their performance remains inconsistent and often declines. To further reveal the unreliability, we start by analyzing the two categories of robust methods mentioned above in the following observations.

Task	Setting	Temperature				
		0	0.3	0.5	0.7	1
Base-9	clean	61.0	60.9	57.5	55.3	46.4
	ina. easy	29.7	<u>28.0</u>	27.2	26.6	21.7
	ina. hard	5.0	<u>5.1</u>	5.5	4.6	5.0
Base-11	clean	34.0	33.8	31.6	29.8	23.9
	irr. easy	21.7	<u>23.1</u>	21.3	23.3	19.1
	irr. hard	<u>17.0</u>	17.5	15.5	14.1	10.7
Sym.(E)	clean	34.2	35.8	<u>35.7</u>	34.6	32.7
	irr. easy	28.6	31.5	<u>29.8</u>	29.1	28.1
	irr. hard	27.0	26.1	<u>26.2</u>	24.0	23.0
Sym.(L)	clean	6.3	8.3	8.9	8.9	9.3
	ina. easy	5.0	7.3	8.6	8.3	7.0
	ina. hard	4.0	6.1	6.3	<u>6.2</u>	6.0

Table 4: Comparing performances of the base model with different temperatures. Sym.(E)/(L) are symbolic tasks.

Task	Setting	#Prompting Examples				
		1	2	3	4	5
Base-9	clean	24.8	38.3	46.4	50.8	50.5
	ina.-easy	17.5	22.2	23.2	25.4	25.6
	ina.-hard	11.3	<u>6.3</u>	6.0	<u>5.7</u>	5.7
Base-11	clean	11.8	20.4	23.9	29.9	32.1
	irr. easy	8.9	15.9	19.1	<u>21.7</u>	26.3
	irr. hard	7.7	10.0	10.7	<u>15.2</u>	16.1
Sym.(E)	clean	18.0	26.5	<u>32.7</u>	39.8	—
	ina.-easy	17.3	23.6	<u>29.1</u>	34.7	—
	ina.-hard	15.0	<u>21.0</u>	22.7	—	—
Sym.(L)	clean	2.7	7.7	9.3	<u>11.3</u>	12.2
	irr. easy	2.3	5.4	7.0	<u>8.8</u>	8.9
	irr. hard	1.9	4.0	<u>6.0</u>	6.3	—

Table 5: Comparing performances of the base model with a varying number of examples ("—" denotes over token limit).

Model	Task	Setting			
		0-shot	clean	irr.	ina.
GPT3.5	Base-9	7.2	46.4	<u>30.3</u>	10.1
	Sym.(E)	8.8	32.7	<u>25.1</u>	26.1
	Com.	40.0	45.7	<u>42.3</u>	33.4
Gemini	Base-9	12.7	88.0	<u>72.3</u>	21.2
	Sym.(E)	9.3	44.5	<u>38.9</u>	36.7
	Com.	42.9	55.6	<u>53.2</u>	33.5
Llama2	Base-9	1.7	4.9	<u>2.9</u>	2.7
	Sym.(E)	4.7	10.1	<u>8.7</u>	9.1
	Com.	35.0	42.3	<u>41.9</u>	40.2
Mixtral	Base-9	3.9	27.5	<u>16.3</u>	3.7
	Sym.(E)	8.3	19.3	<u>17.9</u>	15.1
	Com.	24.2	37.5	<u>34.9</u>	31.1

Table 6: Comparing LLMs with 0-shot, 3-shot clean, and 3-shot medium irrelevant (irr.) / inaccurate (ina.) rationales.

Task	Zero-shot	Few-shot (No Shuffle)	Shuffle Questions x_i	Shuffle Rationales $\mathcal{T}_i$	Shuffle Answers y_i
Math Base-9	7.2	46.4	<u>45.5</u> (0.9%↓)	34.5 (11.9%↓)	35.7 (10.7%↓)
Math Base-11	5.5	<u>23.9</u>	24.8 (0.9%↑)	21.6 (2.3%↓)	21.1 (11.7%↓)
Symbolic Equal	8.8	<u>32.7</u>	<u>32.7</u> (0.0%↓)	32.8 (0.1%↑)	32.3 (0.4%↓)
Symbolic Longer	0.0	9.2	<u>7.0</u> (2.2%↓)	6.2 (3.0%↓)	6.3 (2.9%↓)
Commonsense	40.0	45.7	38.7 (7.0%↓)	39.7 (6.0%↓)	<u>39.8</u> (5.9%↓)

Table 7: Performance (in accuracy%) on NoRa dataset under different few-shot shuffle configurations.

Observation 4.1. Self-correction methods perform poorly on most tasks with noisy rationales. Therein, ISC [29] and SP [89] rely on the inherent capabilities of LLMs to enhance the quality of generated responses. However, in the absence of external feedback, the model’s self-correction ability in reasoning tasks is limited, often resulting in the miscorrection of the given content (see Tab. 12). SP can only slightly improve the accuracy of commonsense tasks, while ISC performs unsatisfactorily across all tasks. As can be seen from Tab. 3, these methods perform even worse than the base model.

Observation 4.2. Self-consistency methods can improve robustness without true denoising. Two self-consistency approaches, SM [62] and SD [102], are originally proposed to address Noisy-Q issues. When applied to our Noisy-R scenarios, they tend to easily disrupt the intrinsic logical coherence within the thought chain. Although these methods utilizing smooth strategies (*e.g.*, random smoothing or masking) perform well on the commonsense dataset, they can hardly handle the more difficult reasoning tasks and even degenerate close to 0%, *e.g.*, in the Symbolic Longer task. Another method, SC [83], performs better than the base model in all tasks, improving both clean and noisy reasoning performance. However, SC does not conduct explicit denoising on rationales during its reasoning procedure. In addition, SC also requires a high computation cost (refer to Appendix F.2).

Besides these methods, next, we analyze LLMs’ *intrinsic* properties under noisy rationales as follows.

Observation 4.3. Adjusting model temperature can help reasoning under noisy rationales. In Tab. 4, we evaluate the base LLM using different temperatures on 3-shot demonstrations. Overall, reducing temperature can enhance the model’s accuracy under both noisy and clean rationale reasoning, compared to the default temperature of 1. However, the relationship between temperature and accuracy is not linear for noisy reasoning; instead, there are multiple peaks in accuracy within the temperature range of 0 to 1. Additionally, it is found that excessively low temperatures (*e.g.*, 0) tend to result in verbose and repeated responses, which cause the model to exceed token limits up to 30% in symbolic tasks where the length of expected answers is quite variable among different questions.

Observation 4.4. Prompting with more noisy examples boosts reasoning accuracy on most tasks. In Tab. 5, we evaluate the model using different numbers of exemplars while keeping the temperature at 1. In general, the LLM’s accuracy will still improve as the number of noisy examples increases in the clean and noisy settings. However, it should be noted that in tasks with high-level noise from NoRa-Math, increasing prompting examples can degenerate accuracy. For example, in the base-9 inaccurate-hard task, prompting with noisy rationales is even worse than the 0-shot accuracy of 7.2%. Further, we provide a deeper analysis of increasing the number of noisy examples in Appendix F.6.